

Lecture 8: Doing the Right Thing

HUM 4441: Engineering Ethics

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The Citicorp Center Case

In the early 1970s, Citicorp planned a new Manhattan headquarters on a block partly occupied by a church. To keep the church at street level, architect Hugh Stubbins and structural engineer William LeMessurier set the 59-story tower on four massive columns placed at the middle of each side, not the corners, with one corner cantilevered over the rebuilt church below.

LeMessurier designed a unique system of 48 chevron-shaped steel braces to carry the wind loads this arrangement created. The building met every applicable code and standard.

An unusual structural problem produced an unusually elegant solution.



Citicorp Center, resting on four stilts.
Fleddermann, Ch. 8, p. 148

A Question That Would Not Go Away

Four years after the tower opened, a question from an engineering student led LeMessurier to check his calculations again. The new numbers showed that under some wind conditions, the braces had to withstand forces about 40% larger than he had originally assumed.

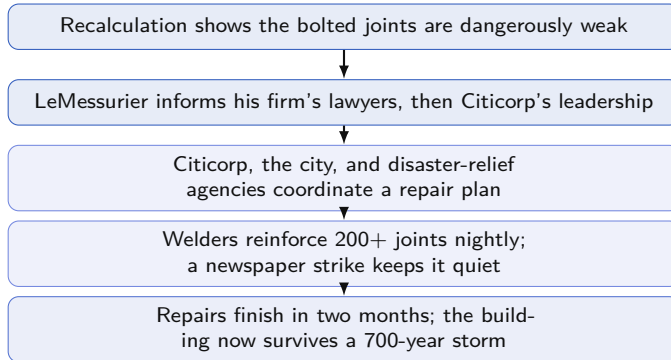
That alone would not have been dangerous. But weeks earlier he had learned that, during construction, the braces' welded joints had been quietly changed to bolted joints, with his own firm's approval. Testing showed the bolted joints would be dangerously weak in a strong storm — and records showed New York could expect one strong enough to tear them apart about once every 16 years.

A structural engineer had just found a hidden flaw in his own celebrated design — one that thousands of people worked and lived above.

Decide before continuing

- Who does LeMessurier tell first: his firm's lawyers, Citicorp, or the city?
- Does it matter that the flaw was introduced by someone else's decision, not his own design?
- What is at risk in every extra day before anyone is told?

From Discovery to Repair



Disclosure did not stay LeMessurier's decision alone — it became a coordinated response across the firm, the client, the city, and emergency planners.

The Outcome

Citicorp did not sue until after the repairs were finished, then settled for \$2 million — the limit of LeMessurier's malpractice insurance. The repairs themselves cost more than \$8 million, against a \$175 million building.

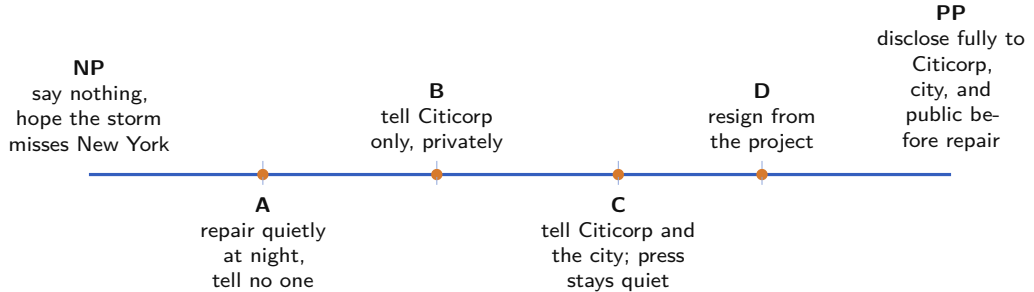
"You have a social obligation. In return for getting a license and being regarded with respect, you're supposed to be self-sacrificing and look beyond the interests of yourself and your client to society as a whole. And the most wonderful part of my story is that when I did it, nothing bad happened."

— William LeMessurier

Coming forward and cooperating did not destroy LeMessurier's reputation — it enhanced it.

Fleddermann, Ch. 8, p. 150

Apply Line Drawing: LeMessurier's Alternatives



Line-drawing task

Option C is close to what LeMessurier actually did. Place it on the line, then decide: was there a more defensible option further toward PP that was still practical, given the risk of panic?

From an Engineer's Courage to a Company's Choice

LeMessurier's decision was personal: one engineer, one building, one moment of disclosure. The next case asks the same question of an entire company — when a safety advance is discovered, does a business have to share it, even when sharing costs it money?

The same duty — put public safety ahead of self-interest — can apply at the scale of a firm, not just an individual.

The Sealed Beam Headlight Case: A Deadly Statistic

In the early days of the automobile, headlamps were unreliable and dim. By 1933, 24 million vehicles were on U.S. roads, with over 31,000 fatalities and 1 million injuries a year. In 1920, 35% of fatalities happened at night; by 1933 that had risen to 56%.

“Visibility has become the weak link in traffic safety, and, as illuminating engineers, we must assume at least a portion of the responsibility in the improvement of traffic hazards.”
— Paul Goodell, 1935

Public safety was paramount here decades before it was written into any code of ethics.

Fleddermann, Ch. 8, p. 150

Why Headlights Failed

Tarnished reflectors

Silvered-brass reflectors tarnished quickly; light output could drop 60% within six months, and owners rarely polished them.

Misaligned, dim bulbs

The filament had to sit at the optical focus within a narrow tolerance, and even a correctly aligned bulb was too dim for highway speed.

By the mid-1930s, decades of small improvements had reached the limit of what the traditional design could deliver.

Fleddermann, Ch. 8, pp. 150–151

Weighing the Alternatives

Light the highways

Enormous upfront and running costs to illuminate thousands of miles, whether or not cars were present.

Restrict night driving

Lower speed limits or limit driving hours; imposes a real cost on every driver, every night.

Fine poor maintenance

Punishes owners after the fact instead of fixing the underlying design.

Every alternative traded cost against benefit; none matched what a better headlight could deliver directly.

Fleddermann, Ch. 8, p. 151

Engineering the Sealed Beam

Two separate GE innovations came together. Daniel Wright's glass-sealing method for movie-projector bulbs — using borosilicate glass with a low thermal-expansion rate — stopped bright bulbs from cracking at the electrical leads. Corning Glass Works then found a way to shape an aluminized glass reflector accurately enough to seal it against the outside environment, ending the tarnishing problem for good.

Two unrelated engineering advances, combined, solved a problem the industry had failed to fix for thirty years.



A sealed-beam headlamp, standardized in 1939–40: bulb, reflector, and lens fused into one sealed unit.
Fleddermann, Ch. 8, pp. 150–151

The Ethical Choice: Share or Protect?

The new design made the headlight a single sealed unit — which threatened to put GE's own customers, the headlamp manufacturers, out of business. GE could have kept the technology proprietary.

Instead, GE demonstrated the design to those same customers, to the Society of Automotive Engineers, and to Ford and GM; let manufacturers consult its own engineers on the design; and worked with regulators so the new headlight would become both accepted practice and a legal requirement as quickly as possible.

GE treated a public-safety advance as something to share industry-wide, not something to defend as a competitive edge.

Fleddermann, Ch. 8, pp. 151–152

Decide: Whose Interest Comes First?

GE's sealed beam threatened to end three of its own customers' businesses — Guide Lamp, C.M. Hall, and Corcoran Brown all made headlamp housings that the new one-piece design made obsolete.

Make a defensible judgment:

- Did GE owe those customers anything, given what the new design would do to their business?
- Is a company's duty to public safety strong enough to require sharing an advantage that costs it, and its partners, money?
- Would the NSPE's rule that safety is paramount over client and employer interests reach this far?

Fleddermann, Ch. 8, p. 152

A Modern Echo: Daytime Running Lamps

In 2001, General Motors petitioned NHTSA to require daytime running lamps — low-intensity headlights that stay on whenever the vehicle is running — on every vehicle sold in the United States.

- NHTSA data: 28% fewer pedestrian fatalities with daytime running lamps.
- GM's own data: 5% fewer accidents.
- Estimated cost: \$20–\$40 per vehicle; already required across much of Europe.
- The U.S. federal government has still not mandated it.

Sixty years apart, two companies reached the same conclusion: some safety advances are worth pushing industry-wide, even at a cost.

Fleddermann, Ch. 8, pp. 152–153

When Companies Disagree About What “Safe” Means

The sealed beam and daytime-lamp cases show engineers sharing a safety advance everyone agreed on. The next case is harder: two respected organizations test the same cars for safety and disagree about the results.

When safety data conflicts, meeting the legal minimum may no longer be enough to call a design responsible.

Two Ways to Crash a Car

	NHTSA	IIHS
Who	Federal regulator, since 1979	Nonprofit funded by auto insurers, since 1995
Test	Full frontal crash into a rigid barrier at 35 mph	Offset frontal crash into a barrier that partially deforms
Status	Sets binding regulations	Publishes recommendations, not legally binding

Some vehicles earning NHTSA's top rating scored the lowest possible IIHS rating — the two tests reveal genuinely different risks.

Fleddermann, Ch. 8, pp. 153–154

The Engineer's Dilemma

Engineers must design to meet NHTSA's binding regulations. IIHS results are not binding at all — but they say something real about how a car behaves in the kind of crash that is actually most common: an offset, not a full frontal, collision.

Decide: An engineer at an automaker learns that a model comfortably meets NHTSA's standard but would score IIHS's lowest rating. What should they recommend to management, and does the cost of the fix change your answer?

Legal compliance is a floor, not a ceiling, when better safety information already exists.

Fleddermann, Ch. 8, p. 154

Avoiding Impediments to Ethical Behavior

Many engineering ethics problems have an obvious right answer, yet engineers do not always choose it. Three excuses come up again and again.

"It's not my problem"

Someone else's mistake, someone else's decision.

"If I don't, someone else will"

Refusing does not stop the technology from being built.

"I can't foresee everything"

No one can predict every consequence of a new design.

Each excuse contains some truth — which is exactly what makes it easy to hide behind.

Fleddermann, Ch. 8, p. 153

Testing the First Two Excuses

“It’s not my problem”

- Lawsuits and redesigns after an unsafe product raise prices for everyone who buys it.
- Injuries raise health-insurance premiums for everyone.
- Contract fraud is made up by taxpayers.

It really is everyone’s problem.

“If I don’t, someone else will”

- Often true — you are rarely the only one working on a given idea.
- But the question is not whether it gets built; it is whether you want to be first to build something harmful.

Being first to notoriety is not a distinction worth having.

Fleddermann, Ch. 8, p. 153

Building In Foresight

"I can't foresee everything" is also true — but engineering is a creative process, and foresight can be built into that process rather than left to chance.

- Design around problems identified in advance, and work with regulators before a new technology ships.
- Treat ethical considerations as part of the design process, not an afterthought.
- Accept that some designs simply should not be built.

If genuine foresight still misses a consequence, that is different from negligence — the right response is to work afterward to make things right, as LeMessurier did.

Fleddermann, Ch. 8, pp. 153–155

Decide: Which Excuse Is This?

A product manager tells your team: *"If we don't ship the location-tracking feature by Friday, our competitor will ship theirs first."*

- Which of the three excuses is being used here?
- What would building foresight into the design process have looked like earlier in the project, before the deadline pressure existed?
- Is there a version of this feature that ships Friday without the ethical shortcut?

Fleddermann, Ch. 8, p. 153

Summary

- LeMessurier's disclosure shows that coming forward about a mistake can protect both public safety and professional reputation.
- GE's sealed beam and GM's daytime running lamps show companies choosing to share a safety advance industry-wide, even at a competitive cost.
- NHTSA and IIHS crash testing shows that meeting the legal minimum is not always the same as designing responsibly.
- "Not my problem," "someone else will," and "I can't foresee everything" are each partly true and each an excuse for inaction.

Reference

Charles B. Fleddermann, *Engineering Ethics*, Chapter 8, “Doing the Right Thing.”